

André Pascoal Bento

Platform Engineer @ Mollie | Assistant Professor @ University of Coimbra | PhD in Informatics Engineering

✉ andre.pascoal.bento@gmail.com ☎ (+351) 910 349 466 📍 Address: Coimbra, Portugal.
🌐 in andre-bento 📄 CISUC Profile 🏆 Scholar 📧 andrepbento

Communication skills: Portuguese (native), English (C1 - advanced), French (B1 - intermediate), Spanish (B1 - intermediate).

About Platform Engineer at Mollie with a Ph.D. in Informatics Engineering from the University of Coimbra, where I also teach as an invited professor. **My academic and industry work focus on optimizing the availability and resource efficiency of cloud services, blending practical engineering with research-driven insights.** I hold a M.Sc. from the University of Coimbra, and a B.Sc. from the Coimbra Institute of Engineering. Passionate about distributed systems and cloud infrastructure, I continuously seek opportunities to build reliable, scalable systems and deepen my expertise. In my free time, outside of work, I enjoy swimming, cycling, exploring nature, and playing the guitar.

Experience

Aug 2025 - Present, Platform Engineer, Mollie

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Sep 2021 - Present, Invited Professor, University of Coimbra

Teaching practical laboratory classes on Distributed Systems (B.Sc.) and Systems Integration (M.Sc.) in Informatics Engineering. Topics in Distributed Systems include multi-threading, parallel programming, RMI/RPC, REST services (Spring Boot and FastAPI), and WebSockets. Systems Integration topics cover microservices, service-oriented architectures, integration patterns, data serialization formats (XML, JSON, and Protocol Buffers), Reactive REST APIs and event-driven systems (Apache Kafka).

Sep 2019 - Present, Researcher, CISUC - Centre for Informatics and Systems of the University of Coimbra

Researching optimization techniques and root-cause analysis to improve availability and resource utilization of cloud services. Developing solutions using Docker, Kubernetes, Helm, Terraform, Ansible, CI/CD and AWS, alongside service mesh (Istio), observability tools (Grafana, Prometheus, and Jaeger) and data analysis using Pandas, NumPy, and SciPy. Utilizing Python, Java, and Go as programming languages and Git as a version control system.

Sep 2018 - Jul 2019, Research Intern, CISUC - Centre for Informatics and Systems of the University of Coimbra

Researched microservices, observability and performance monitoring using metrics, logs, and distributed tracing.

Feb 2017 - Jul 2017, Software Engineer Intern, WIT Software, S.A.

Developed a Mobile Augmented Reality prototype for Android and iOS, implementing digital filters, image manipulation, and user content creation features (e.g., selfies, stickers, photo effects, sound effects, emojis, and drawings).

Oct 2016 - Jun 2017, Scratch Teacher Assistant, CASPAE 10

Taught problem-solving using Scratch programming to primary school children in 3rd and 4th grades.

Nov 2015 - Mar 2016, Math Applied to Engineer Teacher – Volunteer, CeAMatE

Taught mathematics to pre-degree and engineering students.

May 2014 - Jul 2014 and May 2013 - Jun 2013, Accountant Technician Intern, Caixa de Crédito Agrícola Mútuo de Mira - C.R.L.

Assisted with bank accounting, organization, and daily operations.

Education

2019 - 2025, Ph.D. in Informatics Engineering, University of Coimbra

Thesis: Optimizing Availability and Resource Utilization of Cloud Services.

2017 - 2019, M.Sc. in Informatics Engineering, University of Coimbra

Thesis: Observing and Controlling Performance in Microservices.

2014 - 2017, B.Sc. in Informatics Engineering, Coimbra Institute of Engineering (ISEC)

Grants and Projects

- **Dec 2021 - Jul 2025, Ph.D. grant** - Foundation for Science and Technology (FCT) grant number BD.06012.2021.
- **2023 - 2025, ROAR-NET** - Randomised Optimisation Algorithms Research Network (CA22137). Funded by COST.
- **2019 - 2022, AESOP** - Autonomic Service Operation. Funded by P2020-31/SI/2017, No. 040004.

Publications

1. **André Bento**, Filipe Araujo, Luís Paquete, and Raul Barbosa. *Optimal Scaling of Cloud Services*. Submitted to an International Journal.
2. **André Bento**, Filipe Araujo, and Raul Barbosa. *Cost-availability aware scaling: Towards optimal scaling of cloud services*. Journal of Grid Computing, 21(4):80, 2023.
3. Gonalo Baptista, Jaime Correia, **André Bento**, Joao Soares, Antonio Ferreira, Joao Duraes, Raul Barbosa, and Filipe Araujo. *Defektor: An extensible tool for fault injection campaign management in microservice systems*. In Proceedings of the 38th ACM/SIGAPP Symposium on Applied Computing, SAC'23, page 184-187, New York, NY, USA, 2023.
4. Stanley Lima, Filipe Araujo, Miguel de Oliveira Guerreiro, Jaime Correia, **André Bento**, and Raul Barbosa. *Efficient causal access in geo-replicated storage systems*. Journal of Grid Computing, 21(1):8, 2023.
5. **André Bento**, Joao Soares, Ant3nio Ferreira, Joao Duraes, Jos3 Ferreira, Rita Carreira, Filipe Araujo, and Raul Barbosa. *Bi-objective optimization of availability and cost for cloud services*. In 2022 IEEE 21st International Symposium on Network Computing and Applications (NCA), volume 21, pages 45-53. IEEE, 2022.
6. **André Bento**, Jaime Correia, Joao Duraes, Jo3o Soares, Lu3s Ribeiro, Ant3nio Ferreira, Rita Carreira, Filipe Araujo, and Raul Barbosa. *A layered framework for root cause diagnosis of microservices*. In 2021 IEEE 20th International Symposium on Network Computing and Applications (NCA), pages 1-8. IEEE, 2021.
7. Jo3o Tom3s, **André Bento**, Jo3o Soares, Lu3s Ribeiro, Ant3nio Ferreira, Rita Carreira, Filipe Araujo, and Raul Barbosa. *Autonomic service operation for cloud applications: Safe actuation and risk management*. In Dependable Computing-EDCC 2021 Workshops: DREAMS, DSOGR, SERENE 2021, Munich, Germany, September 13, 2021, Proceedings 17, pages 39-46. Springer, 2021.
8. Sara Silva, Jaime Correia, **André Bento**, Filipe Araujo, and Raul Barbosa. *μViz: Visualization of microservices*. In 2021 25th International Conference Information Visualisation (IV), pages 120-128. IEEE, 2021.

9. **Andre Bento**, Jaime Correia, Ricardo Filipe, Filipe Araujo, and Jorge Cardoso. *Automated analysis of distributed tracing: Challenges and research directions*. Journal of Grid Computing, 19:1-15, 2021.